

Rabia Samad Afridi

EMR 19, Dubai, UAE | +971 5081188657 | rabiasamad895@gmail.com
<https://www.linkedin.com/in/rabia-samad-3a225b163/>

AI Engineer with a strong problem-solving mindset, specializing in LLMs and prompt engineering for real-world applications. Experienced in building AI voice agents, RAG systems, TTS and video generation models, and scalable automation solutions.

TECHNICAL SKILLS

Programming Languages: Python,

Frameworks/Libraries: TensorFlow, PyTorch, LangChain, Lang graph

ML & AI: Machine Learning, Computer Vision, Prompt Engineering, LLM Function Calling, Routing, Memory Management

APIs & Development: Flask API

Cloud Platforms: AWS (EC2, S3)

Tools & Utilities: LangChain, Lang graph, N8N, Heygen, vapi, Zapier

PROFESSIONAL EXPERIENCES

Project Lead | Smart Forum

May 2025 – Present

Project: Dentally Assist (AI Voice Agent /Receptionists)

- Defined project scope, developed project charter, and established timelines to ensure structured execution and delivery.
- Led a cross-functional team of 9 (Developers, UI/UX ,QA, DevOps) to build an AI-powered voice assistant for dental clinic operations.
- Designed an AI voice agent for scheduling appointments in Dentally using the Dentally API.
- Currently developing a dashboard for human receptionists to monitor call logs and manage interactions efficiently.

Project: Curio (Mobile Application – Early Autism Childhood Development)

- Awarded Star Performer award (4th quarter 2025) for outstanding performance.
- Led an agile cross-functional team of 15 (Developers, Designers, QA, DevOps, Domain Experts) to deliver, deploy, and launch a mobile application focused on autism early detection, learning, and development.
- Owned end-to-end product development and AI integration: collaborated with psychologists and therapists to ensure content alignment; developed RAG-based chatbot for parental guidance; built AI-powered therapy modules and interactive assessment tools for autism early detection; managed project planning, timelines, and delivery milestones.
- Launched the application on the Google Play Store, achieving **500+** downloads.
- Leading ongoing iOS application development, with the app currently in progress for an upcoming launch on the Apple App Store to expand platform reach.

AI Engineer | Smart Forum

August 2023 – May 2025

- Led end-to-end development of an AI-driven media platform, R&D on open-source models (LLMs, TTS, lip-sync) to evaluate feasibility, enable local deployment, and validate system workflows.
- Collected and prepared data to fine-tune lip-sync model, enhancing synchronization and overall output quality by 60% and reduce the response time by almost 50%.
- Applied advanced prompt engineering (zero shot +role prompting) techniques to control LLM behavior, enabling automated generation of news articles and scripts in a consistent “news caster” style and tone.
- Designed and implemented automation workflows to generate, process, and publish content, including one-click deployment of articles directly to client websites via API integration.
- Platform is live and actively used by clients, reducing news and article generation time by 80%.

RESEARCH EXPERIENCE

Classification of Liver Disease Using Neural Networks: [JSN-SR04T (Ultrasonic sensor)]

- Detecting early-stage liver cancer using Ultrasonic sensor and training the system using Neural Networks.

Segmentation of hands from image

- Using computer vision technique like Hough Transform and Harris/SIFT detection on images to detect hands center.

ACHEIVEMENTS

- Placed in Dean's List of Honor for securing SGPA greater than 3.5 in spring'18 and fall'18 semesters.
- Awarded with Bronze Medal for outstanding performance in Fall'17 semesters.

CERTIFICATIONS

- Prompt Engineering with Llama 2 & 3 – DeepLearning.AI
- Functions, Tools and Agents with LangChain – DeepLearning.AI
- LangChain Chat with Your Data – DeepLearning.AI
- Generative AI with Large Language Models – DeepLearning.AI
- Introducing Multimodal Llama 3.2 – DeepLearning.AI
- LangChain for LLM Application Development – DeepLearning.AI
- Multi AI Agent Systems with CrewAI – CrewAI
- Introduction to AI and Machine Learning on Google Cloud – Google Cloud Training Online

EDUCATION

Master's degree from NUST

September 2021 – January 2024

Coursework: Machine learning, Deep learning, Computer Vision, Computer Architecture, Mobile Robotics, Stochastic systems.

Bachelor's degree from **National University of Computer and Emerging Sciences**

August 2015 – June 2019

Coursework: Computer Architecture, Intro to Robotics, Data Communication and Networks, Embedded Systems, Digital Signal Processing, Analogue and Digital Communication.